

MERIDIAN-1 DATA CENTRE, HALL A

Project No. MER-1-2026 · Issued for Construction · Revision 0 · 2026-01-15

SECTION 26 33 53 — STATIC UNINTERRUPTIBLE POWER SUPPLY (UPS)

26 33 53 Part 1 — GENERAL

1.1 SUMMARY. This Section specifies double-conversion static UPS systems serving the Hall A critical bus, including battery systems, battery cabinets, and accessories. Equipment tags UPS-A1 through UPS-A4.

1.2 REFERENCES. IEC 62040-3, IEEE 485, IS 1893 (Part 4), NFPA 70E.

1.3 SUBMITTALS. Product data, factory test reports, seismic anchorage calculations, and battery sizing calculations per IEEE 485. The Contractor shall identify any deviation from this Section on the transmittal; undeclared deviations discovered during review shall be grounds for rejection without further evaluation.

26 33 53 Part 2 — PRODUCTS

2.1 EFFICIENCY. UPS efficiency in double-conversion (VFI) mode shall be not less than 97.0 percent at 100 percent rated load, verified per IEC 62040-3. Eco-mode or bypass-mode efficiency figures shall not be substituted for VFI figures.

2.2 BATTERY AUTONOMY. The battery system shall provide not less than 10 minutes autonomy at full rated UPS load, at end-of-life capacity, at 25 degrees C.

2.3 INPUT HARMONICS. Input current total harmonic distortion (THDi) shall not exceed 5 percent at 100 percent load. Compliance shall be achieved by the UPS rectifier alone, without external or optional harmonic filter modules.

2.4 OVERLOAD. The inverter shall sustain 125 percent load for 10 minutes and 150 percent load for 60 seconds.

2.5 PARALLELING. Units shall be capable of paralleling in systems of not less than six units in N+1 configuration without a centralised static switch.

2.6 BATTERY CABINETS. Battery cabinets shall be seismically anchored per IS 1893 (Part 4). Anchor bolt torque records shall be submitted as a quality record prior to energisation.

2.7 EFFICIENCY VERIFICATION. Factory witness test per IEC 62040-3; see also any Addendum modifying the referenced edition.

26 33 53 Part 3 — EXECUTION

3.1 FACTORY TESTS. Full-load heat run, efficiency verification at 25/50/75/100 percent load, transfer tests, and battery discharge test witnessed by the Owner's representative.

3.2 SITE ACCEPTANCE. Level 4 component verification and Level 5 integrated systems test participation per Section 01 91 13.